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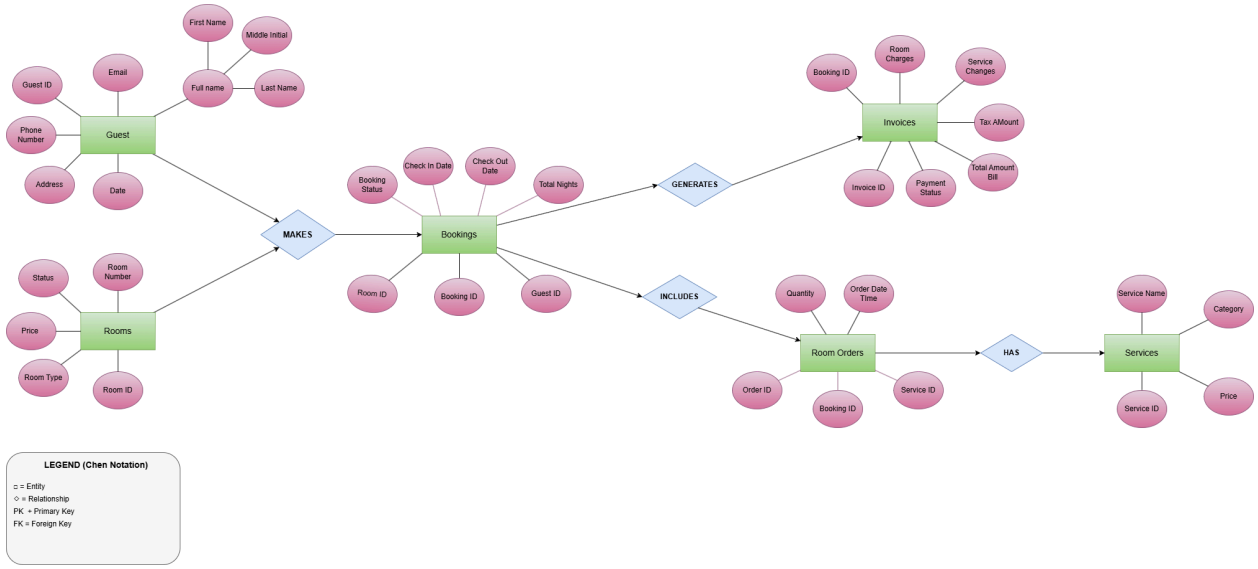
# PROJECT #9 – HOTEL CONCIERGE & BILLING SYSTEM

## Project Overview

Core Functionality:

- Manage guest check-ins and check-outs
- Track room service orders and additional charges
- Consolidate multiple charges into a single invoice using complex SQL joins
- Automate billing calculations via Python backend

## Chen Diagram



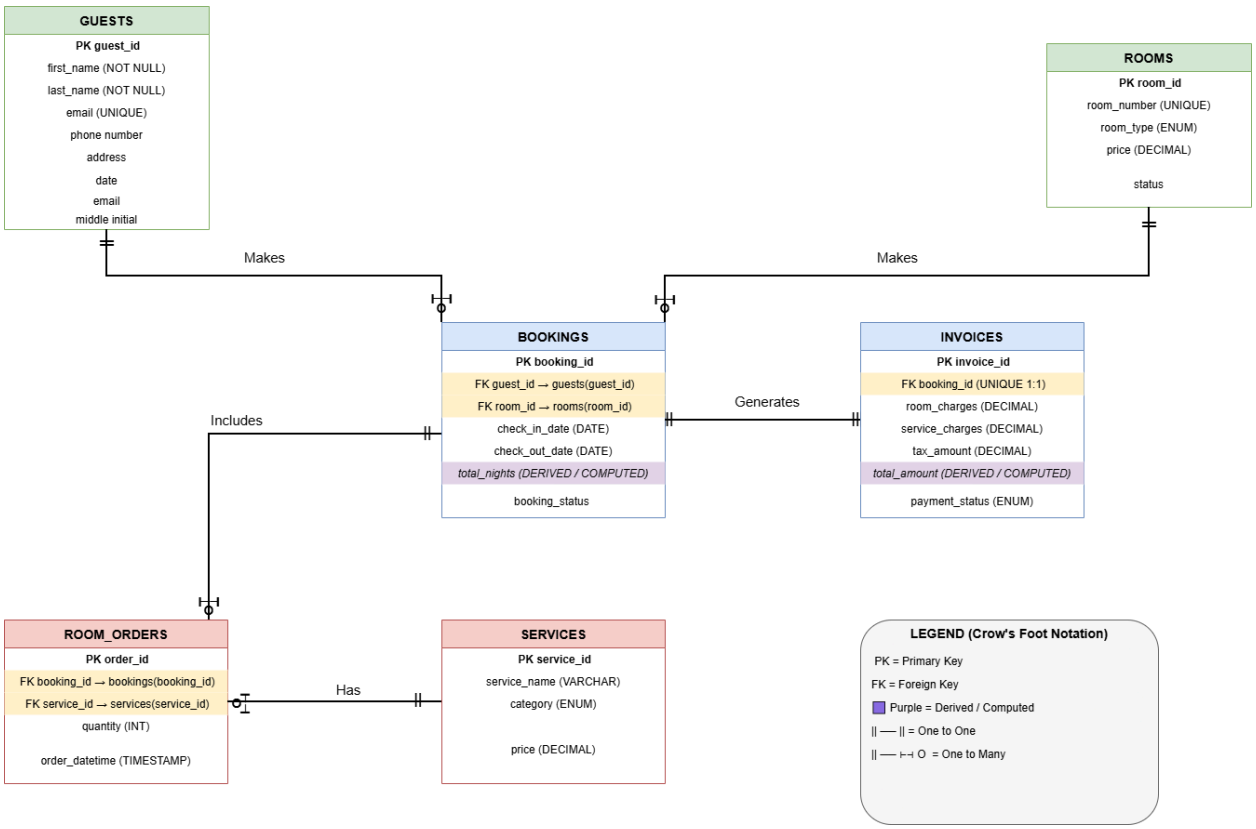

The diagram shows six (6) entities connected by five (5) relationships.

**ENTITIES (6 total):**
1. GUESTS - Stores personal information of hotel customers
2. ROOMS - Contains room inventory with types and pricing
3. BOOKINGS - Links guests to rooms with check-in/out dates
4. SERVICES - Menu of available add-on services (restaurant, spa, etc.)

5. ROOM_ORDERS - Records service requests made during a stay
6. INVOICES - Generates consolidated bills for each booking

**RELATIONSHIPS (5 total):**
• MAKES (1:M) - One guest makes many bookings; one room has many bookings
• GENERATES (1:1) - One booking generates exactly one invoice
• INCLUDES (1:M) - One booking can include many room orders
• HAS (1:M) - One service can appear on many room orders

# Crow Foot Diagram




The diagram shows six (6) tables connected by five (5) relationships with cardinalities.

**TABLES (6 total):**
1. GUESTS - Stores personal information of hotel customers
2. ROOMS - Contains room inventory with types and pricing
3. BOOKINGS - Links guests to rooms with check-in/out dates
4. SERVICES - Menu of available add-on services (restaurant, spa, etc.)
5. ROOM_ORDERS - Records service requests made during a stay
6. INVOICES - Generates consolidated bills for each booking

**RELATIONSHIPS (5 total):**

- guests → bookings (1:M) - One guest can make many bookings
- rooms → bookings (1:M) - One room can have many bookings over time
- bookings → invoices (1:1) - One booking generates exactly one invoice
- bookings → room_orders (1:M) - One booking can include many service orders
- services → room_orders (1:M) - One service can be ordered many times

# Data Dictionary

| | | | | | |
|---|---|---|---|---|---|
| 1 | TABLE 1: guests | | | | |
| 2 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 3 | guest_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique identifier for each guest |
| 4 | first_name | VARCHAR | NOT NULL | NO | NO | Guest's first name |
| 5 | last_name | VARCHAR | NOT NULL | NO | NO | Guest's last name |
| 6 | email | VARCHAR | UNIQUE, NOT NULL | NO | NO | Guest's email address |
| 7 | phone | VARCHAR | None | NO | NO | Contact phone number |
| 8 | address | TEXT | None | NO | NO | Complete residential address |
| 9 | created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | NO | NO | Registration date and time |
| 10 | | | | | | |
| 11 | TABLE 2: rooms | | | | | |
| 12 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 13 | room_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique room identifier |
| 14 | room_number | VARCHAR | UNIQUE, NOT NULL | NO | NO | Physical room number/name |
| 15 | room_type | ENUM | NOT NULL | NO | NO | Category of room (Standard, Deluxe, Suite, Presidential) |
| 16 | price_per_night | DECIMAL | CHECK > 0, NOT NULL | NO | NO | Cost per night in local currency |
| 17 | status | ENUM | NOT NULL | NO | NO | Current room availability (Available, Occupied, Maintenance, Reserved) |
| 18 | | | | | | |
| 19 | TABLE 3: bookings | | | | | |
| 20 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 21 | booking_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique booking reference |
| 22 | guest_id | INT | FOREIGN KEY, NOT NULL | NO | YES (guests.g | Guest making the booking |
| 23 | room_id | INT | FOREIGN KEY, NOT NULL | NO | YES (rooms.r | Room being booked |
| 24 | check_in_date | DATE | NOT NULL | NO | NO | Arrival date |
| 25 | check_out_date | DATE | NOT NULL | NO | NO | Departure date |
| 26 | total_nights | INT | GENERATED, STORED (DERIVED) | NO | NO | Number of nights (auto-calculated) |
| 27 | booking_status | ENUM | NOT NULL | NO | NO | Current booking state (Active, CheckedOut, Cancelled) |
| 28 | | | | | | |
| 29 | TABLE 4: services | | | | | |
| 30 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 31 | service_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique service item ID |
| 32 | service_name | VARCHAR | NOT NULL | NO | NO | Name of service/product |
| 33 | category | ENUM | NOT NULL | NO | NO | Service department category (Restaurant, Spa, MiniBar, Laundry, Other) |
| 34 | price | DECIMAL | CHECK > 0, NOT NULL | NO | NO | Cost per unit/item |
| 35 | | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| 36 | TABLE 5: room_orders | | | | |
| 37 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 38 | order_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique order reference |
| 39 | booking_id | INT | FOREIGN KEY, NOT NULL | NO | YES (booking | Associated booking |
| 40 | service_id | INT | FOREIGN KEY, NOT NULL | NO | YES (services | Service ordered |
| 41 | quantity | INT | CHECK > 0, NOT NULL | NO | NO | Number of units ordered |
| 42 | order_datetime | DATETIME | DEFAULT CURRENT_TIMESTAMP | NO | NO | When order was placed |
| 43 | | | | | | |
| 44 | TABLE 6: invoices | | | | | |
| 45 | Attribute | Data Type | Constraints | Primary Key | Foreign Key | Description |
| 46 | invoice_id | INT | PRIMARY KEY, AUTO_INCREMENT | YES | NO | Unique invoice number |
| 47 | booking_id | INT | FOREIGN KEY, UNIQUE, NOT NULL | NO | YES (booking | Associated booking (one-to-one) |
| 48 | room_charges | DECIMAL | DEFAULT 0 | NO | NO | Total room cost (nights × price) |
| 49 | service_charges | DECIMAL | DEFAULT 0 | NO | NO | Total of all service orders |
| 50 | tax_amount | DECIMAL | DEFAULT 0 | NO | NO | Tax calculated (12% of subtotal) |
| 51 | total_amount | DECIMAL | GENERATED, STORED (DERIVED) | NO | NO | Grand total (auto-calculated) |
| 52 | payment_status | ENUM | NOT NULL | NO | NO | Payment tracking (Unpaid, Paid, PartiallyPaid) |
| 53 | invoice_date | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | NO | NO | Bill generation date |
| 54 | | | | | | |